

PROJECT PROPOSAL

Musify — Desktop Music Player Application

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Roll Number	2026-CYS-14
Subject	Programming Fundamentals
Submitted To	Hafiz Muhammad Mubashar
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1. Project Title

Musify — A Desktop Music Player Application

2. Problem Statement

Managing and playing local audio files on desktop computers often requires bulky third-party applications or platform-native tools that are difficult to customize. There is a need for a lightweight, visually appealing, and easily extensible music player application that can be built entirely using open-source tools and standard programming concepts.

This project proposes the development of a fully functional desktop music player called Musify, built using Python and the PyQt5 GUI framework, that allows users to organize and play their local audio library in a clean, modern interface.

3. Project Objectives

The objectives of this project are as follows:

- To develop a functional desktop music player using Python and PyQt5
- To demonstrate the application of OOP, inheritance, and encapsulation
- To implement event-driven programming through PyQt5 signals and slots
- To build a visually appealing dark-themed GUI inspired by modern music platforms
- To integrate multimedia playback capabilities using QMediaPlayer
- To handle file system operations including browsing and loading audio files

4. Proposed Features

4.1 Core Features (Phase 1)

- Audio playback for MP3, WAV, and OGG file formats
- Play, pause, stop, next, and previous track controls
- Real-time progress slider with seek functionality
- Volume control with dedicated slider
- Song queue management via a scrollable list
- File selection dialog to add songs from the local filesystem

4.2 UI/UX Features

- Dark background (Spotify-inspired #121212) with green accents (#1DB954)
- Left sidebar with navigation labels and playlist stubs
- Song list table with track number, title, album, and duration columns
- Bottom player bar showing current track info, controls, and sliders
- Dynamic play/pause button icon reflecting actual playback state

5. Technical Approach

5.1 Architecture

The application will follow a single-class architecture where `MusicPlayer` extends `QMainWindow`. All state variables (player, song list, current index, timer) and methods (UI builders, playback logic, event handlers) will be encapsulated within this class.

5.2 Key Technologies

Technology	Justification
Python 3	Readable syntax, rapid development, strong community support
PyQt5	Mature, cross-platform GUI framework with built-in multimedia support
QMediaPlayer	Built-in audio engine supporting all target file formats
QTimer	Reliable periodic updates for progress slider synchronization
QFileDialog	Native file picker for seamless cross-platform file selection

6. Project Timeline

Phase	Duration	Deliverable
Phase 1: Planning	1 week	Requirements gathering, UI wireframe design
Phase 2: UI Development	1 week	Sidebar, main area, bottom bar layout
Phase 3: Core Logic	1 week	Playback engine, song loading, controls
Phase 4: Integration	3 days	Signal-slot wiring, state management
Phase 5: Testing	2 days	Bug fixing, edge cases, UI polish
Phase 6: Submission	1 day	Report, slides, final code submission

7. Expected Outcomes

Upon successful completion of this project, the following outcomes are expected:

- A fully functional Python desktop music player application
- Demonstrated mastery of PyQt5 GUI and multimedia APIs
- A well-structured codebase following OOP design principles
- A professional project report and presentation slides
- Enhanced understanding of event-driven programming in real applications

8. Resources Required

- Python 3.8 or higher
- PyQt5 library (pip install PyQt5)
- A code editor (VS Code, PyCharm, or similar)
- Sample audio files for testing (MP3/WAV/OGG formats)
- Access to official PyQt5 and Qt documentation

9. Conclusion

The Musify Music Player project is a well-scoped, achievable, and educationally valuable project for the Programming Fundamentals course. It directly applies OOP, event-driven programming, GUI design, and multimedia integration in a single cohesive application. The project balances technical depth with visual impact, making it an ideal demonstration of programming competency.

I respectfully request approval to proceed with this project proposal.

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